

CHRISTAL CHEN

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EDUCATION

Princeton University

Princeton, NJ

*Bachelor of Science in Engineering, Electrical and Computer Engineering
Robotics and Cyberphysical Systems Concentration*

May 2028

GPA: 3.71 / 4.00

- **Intended Minor:** Computer Science, Robotics
- **Relevant Coursework:** Information Signals (ECE201), Circuit Design (ECE203), Logic Design (ECE206), Algorithms and Data Structures (COS226), Programming Systems (COS217), Machine Learning (COS324), Linear Algebra (MAT202), Differential Equations (MAE305), Probability (ORF309)
- **Activities:** Chinese Language Association (Vice President), Six14 Christian Dance Company (Publicity Chair), Matriculate (Advising Fellow), Birding Society (Communications VP), Hoagie Club, Robotics, Electric Speedboat

RESEARCH

Princeton Safe Robotics Lab

Nov 2025 – Present

Undergraduate Researcher

- Enhance a gameplay-based safety filter for autonomous robotic systems by implementing multi-scenario rollouts
- Train an agent with reinforcement learning and run simulated safety-validation experiments in PyBullet, experimentally determining the optimal number of rollouts for the safety filter
- Deploy and validate the extended gameplay safety filter on a Unitree Go2 quadruped, conducting simulation-to-hardware transfer experiments to assess real-world safety performance under a 50N adversarial force

AP Research Study

Aug 2022 – May 2023

Primary Researcher

- Conducted an independent study analyzing ecological restoration in Austin, TX, using eBird citizen-science datasets
- Used R and SQL to organize and analyze large distribution datasets, calculating an index to measure habitat change
- Worked with a Global Ecology researcher at the University of Florida to refine methods and validate findings
- Gained experience with the full research process, including submitting a study proposal through an IRB, selecting comparable study areas to maximize insight, and writing a paper with a literature review to contextualize findings

EXPERIENCE

Princeton University Robotics Club

Sep 2024 – Present

Mechanical Designer

- Design a waste-collecting robot in Onshape with a focus on mechanical reliability and modular components
- Assemble 3D-printed frames with motors, servos, sensors, and potentiometers for full system integration
- Work in a collaborative team of 5 to prototype ideas, troubleshoot issues, and iterate on mechanical designs
- Design and wire a custom circuit in collaboration with the electrical team for component integration

Hoagie Club

Sep 2025 – Present

Software Developer - HoagieMail

- Contribute to the development of a mass-emailing platform used by hundreds of student organizations
- Create and debug 2 features (newsletter and email templates) while maintaining organized Git workflows
- Collaborate with 95 developers to brainstorm and plan new app features that improve usability and performance
- Submit detailed pull requests in GitHub to maintain code quality and expand documentation

Princeton Electric Speedboating

Sep 2025 – Present

Embedded Software Engineer

- Develop an ESP32 CAN bus interface to communicate with a battery management system for an electric speedboat
- Implement non-blocking CAN message polling, request/response handling, and data caching for real-time telemetry
- Parse CAN payloads and structure message decoders for reliable data extraction on 6 signals (state of charge, overall cell temperature, individual cell temperatures, motor speed, motor temperature, current usage)

SKILLS

Languages: Python, Verilog, C, Java, R, SQL, JavaScript, HTML/CSS, C++ (Beginner)

Software Tools: Vivado, Git, Onshape CAD, MATLAB, PyBullet (Beginner), MuJoCo (Beginner)

Hardware: Arduino, FPGA (Beginner)

Other Skills: Mandarin Chinese, Adobe Suite, Graphic Design